Mostafa El Bakhti

Research Assistant – PFE 2027 | Software Engineering Student

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SUMMARY

Software engineering student at 1337 Coding School (42 Network), applying for the Research Assistant – PFE 2027 internship at Oracle Morocco R&D Center. Hands-on experience in Python, C, pathfinding algorithms and LLM function calling, with one year of earlier training at SoliCode (2024) in HTML, CSS and JavaScript. Eager to contribute algorithmic problem-solving and debugging to Oracle R&D while learning from industry experts on emerging technologies and AI.

SKILLS

Programming Languages: Python, C, JavaScript
Software Engineering & Algorithms: Software engineering, data structures, algorithm design, pathfinding, sorting algorithms, memory management, debugging
AI & Emerging Technologies: LLM function calling, PyTorch, Hugging Face Transformers, tokenization, structured JSON output
Web Development: HTML, CSS, JavaScript
Tools & Systems: Linux, Debian, system administration, Bash, Git, GitHub, virtual machines

PROFESSIONAL EXPERIENCE

Software Engineering Student | 1337 Coding School (42 Network), Tétouan, Morocco | Current

- Completed 7 software projects and 11 Python modules in Python and C, covering pathfinding, sorting algorithms, maze generation and LLM function calling.
- Collaborated with fellow students through 1337's project-based, peer-learning model to solve algorithmic problems and debug code.
- Configured and administered a Linux environment and managed project code with Git and GitHub.

PROJECTS

Call Me Maybe | *Python, PyTorch, Hugging Face Transformers*

- Prototyped constrained text generation that maps natural-language requests to structured JSON function calls.
- Analyzed logits over a 151,936-token vocabulary and tested next-token masking to restrict the model's output choices.

Fly-in (in progress) | *Python, Pathfinding, Weighted Graphs*

- Developing route planning for drones moving through connected zones with different traversal costs.
- Validating route occupancy constraints for zones and links.

A-Maze-ing | *Python, Maze Algorithms*

- Developed a grid-based maze generator and solver, applying algorithmic reasoning and appropriate data representations.

push_swap | *C, Data Structures*

- Implemented a stack-based sorting program under restricted operations, focused on efficient move sequences.

Libft, ft_printf & get_next_line | *C*

- Built a reusable library of string and memory functions in C, and re-implemented printf and line-by-line file reading from scratch.

Born2beroot | *Linux, Debian, Bash*

- Configured and administered a secured Debian virtual machine.

Python Modules 00-10 | *Python*

- Completed 11 modules covering Python fundamentals through object-oriented design.

EDUCATION

1337 Coding School (42 Network) – Software Engineering Program | Tétouan, Morocco | Current
SoliCode – One-year training program | HTML, CSS, JavaScript | 2024